But that’s exactly what Einstein explained to Bohr—in quantum mechanics, one can know everything about a system and nothing about its individual parts—but Bohr failed to appreciate this fact. I might add that generations of quantum textbooks blithely ignored it.

Everyone knows that quantum mechanics is strange, but I suspect very few people could tell you exactly in what way. This book is a technical course of lectures on quantum mechanics, but it is different than most courses or most textbooks. The focus is on the logical principles and the goal is not to hide the utter strangeness of quantum logic but to bring it out into the light of day.

I remind you that this book is one of several that closely follow my Internet course series, the Theoretical Minimum. My coauthor, Art Friedman, was a student in these courses. The book benefited from the fact that Art was learning the subject and was therefore very sensitive to the issues that might be confusing to the beginner. During the course of writing, we had a lot of fun, and we’ve tried to convey some of that spirit with a bit of humor. If you don’t get it, ignore it.

Leonard Susskind

When I completed my master’s degree in computer science at Stanford, I could not have guessed that I’d return some years later to attend Leonard’s physics lectures. My short “career” in physics ended many years earlier, with the completion of my bachelor’s degree. But my interest in the subject has remained very much alive.

It appears that I have lots of company—the world seems filled with people who are genuinely, deeply interested in physics but whose lives have taken them in different directions. This book is for all of us.

Quantum mechanics can be appreciated, to some degree, on a purely qualitative level. But mathematics is what brings its beauty into sharp focus. We have tried to make this amazing body of work fully accessible to